

Deep Learning for Text Analytics

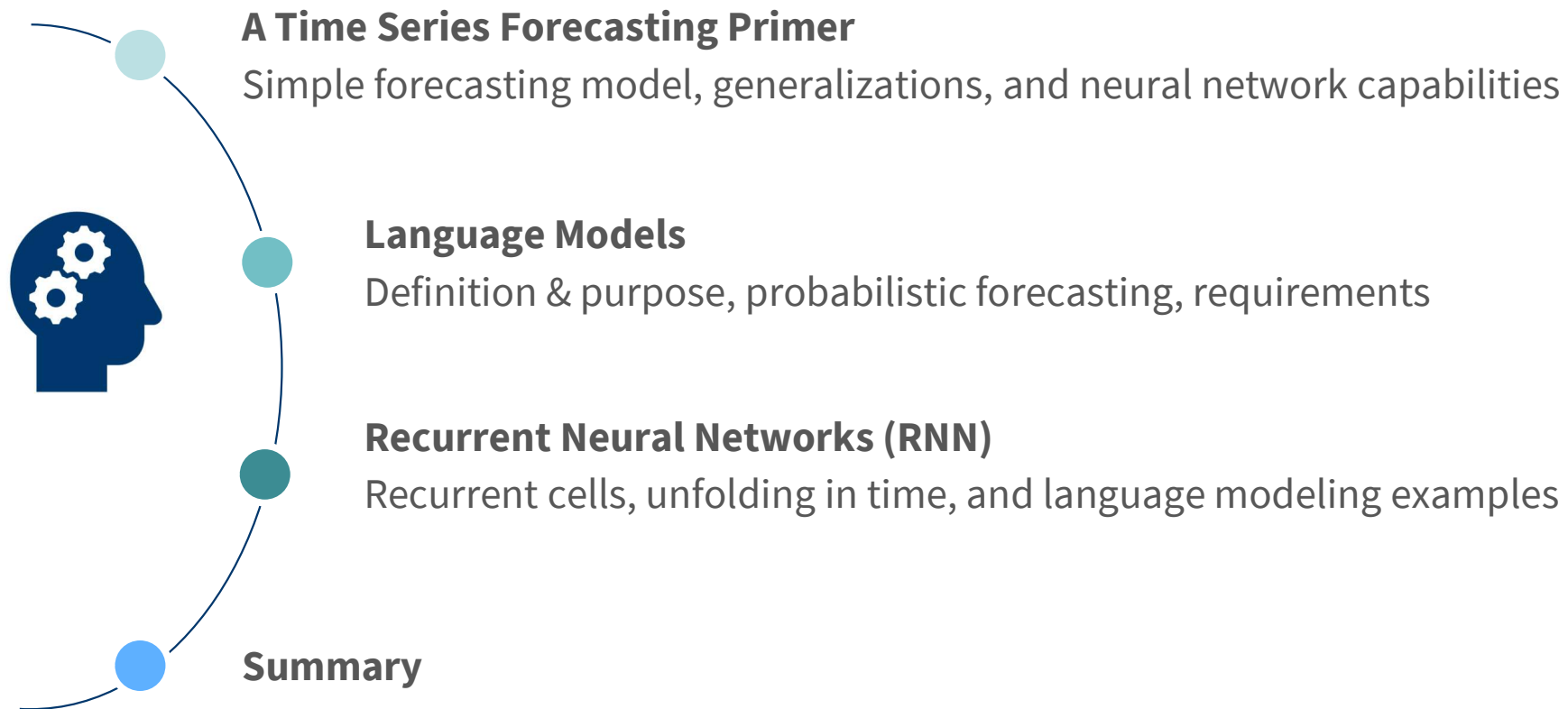
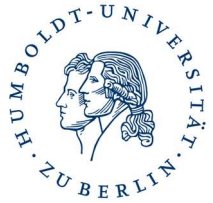


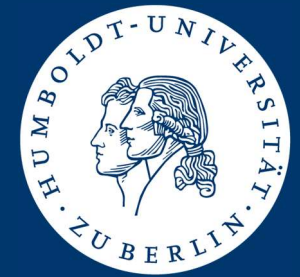
Network Architectures for Sequential Data – Recurrent Neural Networks I

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Agenda





A Time Series Forecasting Primer

Simple forecasting model, generalizations, and neural network capabilities

Time-Ordered – Sequential - Data is Omnipresent

Economic time series, sales data, inventory numbers, ...

■ Time series analysis

- ☐ Test if stationary
- ☐ Test if seasonal
 - Interval
 - Yearly, monthly, ... intra-day
- ☐ Test if trend
 - Additive trend
 - Multiplicative trend

■ Time series forecasting

- ☐ Analysis supports model selection
- ☐ E.g., include seasonal component (e.g., ARIMA vs SARIMA)



German stock index (DAX 40) since 1959

<https://www.finanzen.net/index/dax/seit1959>

A Simple Time Series Forecast Model

Formal representation and corresponding neural network

■ Notation

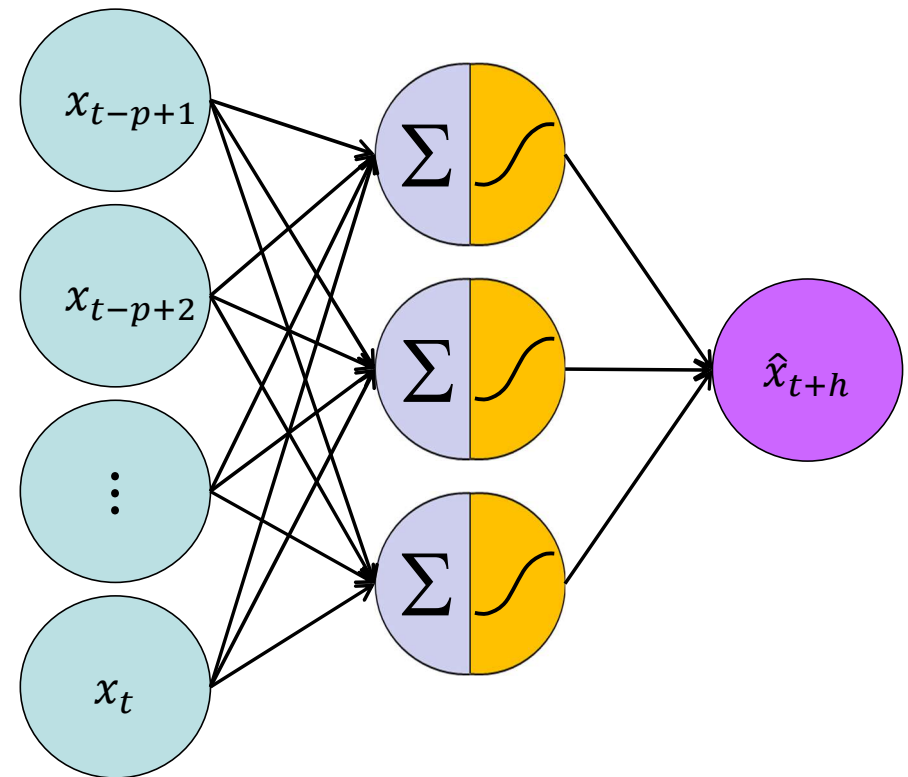
- $x_t \in \mathbb{R}$: observed value at time t
- $\mathbf{x}_{t-p+1:t} = (x_{t-p+1}, x_{t-p+2}, \dots, x_t)$: window of past p observations of x
- $\hat{x}_{t+h}$: point forecast h steps ahead

■ Univariate time series forecasting model, f_w , with parameters w

- $\hat{x}_{t+h} = f_w(\mathbf{x}_{t-p+1:t})$
- f_w could be linear or nonlinear

■ Note

- Autoregressive (AR) models predict one step ahead
- Roll forecast forward to predict multiple steps into the future

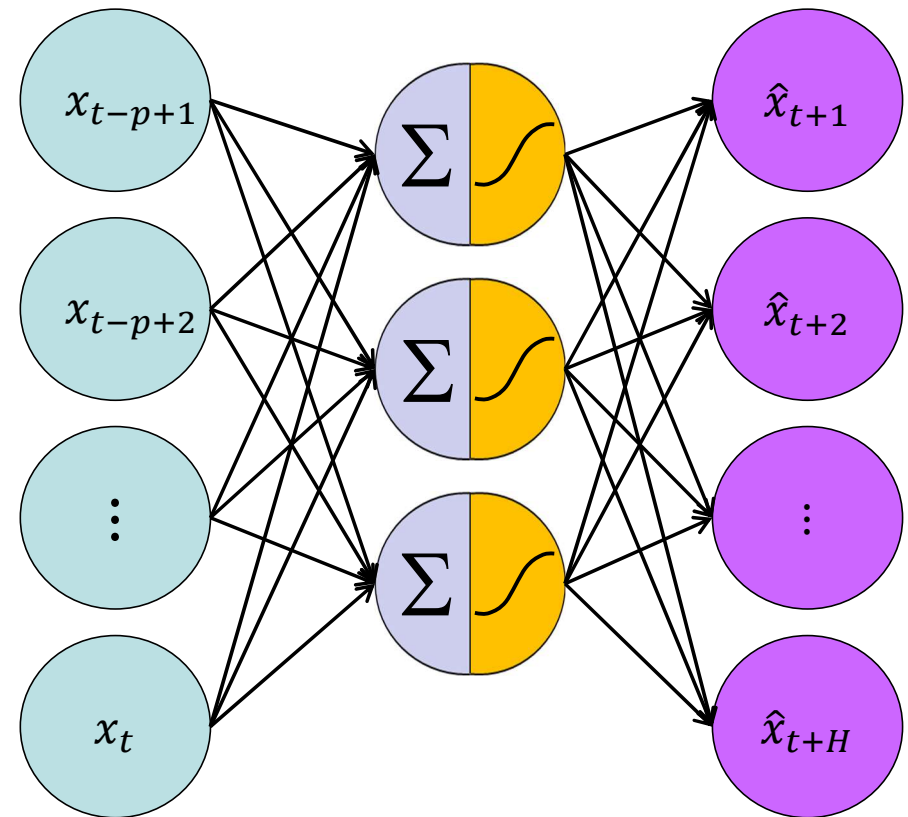


Generalized Time Series Forecasting Models

Univariate multi-step forecasting model

■ Using the same input data, forecast multiple steps into the future

- $\hat{\mathbf{x}}_{t+1:t+H} = f_w(\mathbf{x}_{t-p+1:t})$
- $\hat{\mathbf{x}}_{t+1:t+H} = (\hat{x}_{t+1}, \hat{x}_{t+2}, \dots, \hat{x}_{t+H})$, vector for forecasts for a window of H future periods
- H is also called forecast horizon



Generalized Time Series Forecasting Models

Multivariate forecasting with external features

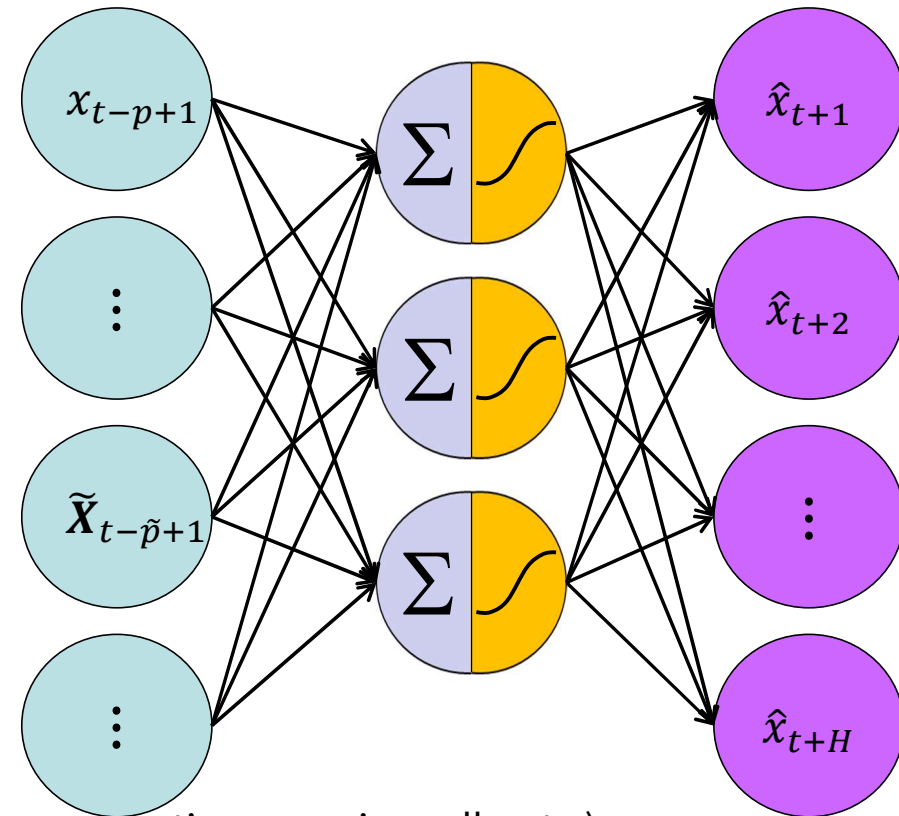
■ Single period formulation:

$$\hat{x}_{t+h} = f_w(x_{t-p+1:t}, \tilde{X}_{t-p+1:t})$$

■ Multi-period formulation (visualized):

$$(\hat{x}_{t+1}, \hat{x}_{t+2}, \dots, \hat{x}_{t+H}) = f_w(x_{t-p+1:t}, \tilde{X}_{t-\tilde{p}+1:t})$$

- $\tilde{X}_{t-\tilde{p}+1:t} \in \mathbb{R}^{\tilde{p} \times \tilde{d}}$ denotes a matrix collecting the values of $\tilde{d}$ external features over $\tilde{p}$ time steps
- The time window over which we gather the external features could differ from p , the window of past realizations of the forecast time series x .
 - This is why we introduce $\tilde{p}$
 - Given the features are external, they could also incorporate future values if these are known (e.g., calendar events, upcoming promotions, earning calls, etc.)
 - Notation would then change to $\tilde{X}_{t-\tilde{p}+1:t+\tilde{H}}$, with $\tilde{H}$ denoting the horizon over which we use future values of $\tilde{X}$



Generalized Time Series Forecasting Models

(Global) Multivariate multi-output forecasting model

■ Single-period forecasting w/o external features: $\hat{x}_{t+h} = f_w(X_{t-p+1:t})$

- $X_{t-p+1:t} \in \mathbb{R}^{p \times d}$ matrix of p past feature values of a set of d time series
- $\hat{x}_{t+h} \in \mathbb{R}^d$ vector of one-step-ahead forecasts of the d time series

■ Multi-period forecasting w/o external features: $\hat{X}_{t+1:t+H} = f_w(X_{t-p+1:t})$

- $X_{t-p+1:t} \in \mathbb{R}^{p \times d}$ matrix of p past feature values of a set of d time series
- $\hat{X}_{t+1:t+H} \in \mathbb{R}^{H \times d}$ matrix all d time series forecasts for the periods $t + 1, t + 2, \dots, t + H$

■ Multi-period forecasting with external features: $\hat{X}_{t+1:t+H} = f_w(X_{t-p+1:t}, \tilde{X}_{t-\tilde{p}+1:t})$

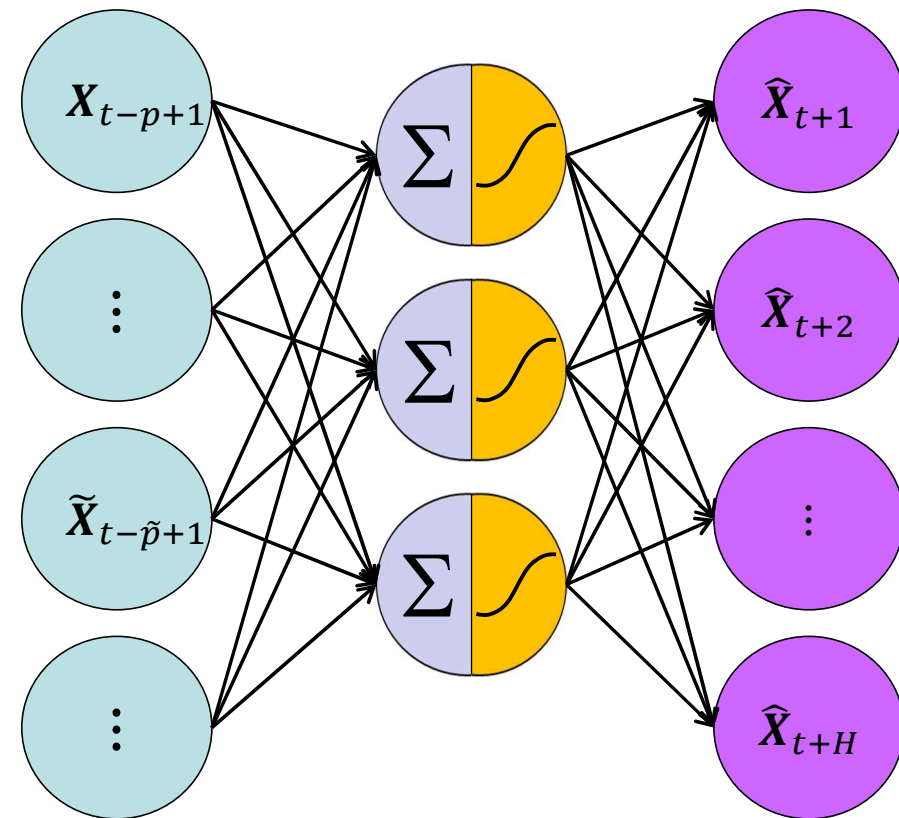
- $X_{t-p+1:t} \in \mathbb{R}^{p \times d}$ matrix of p past feature values of a set of d time series
- $\tilde{X}_{t-\tilde{p}+1:t} \in \mathbb{R}^{p \times \tilde{d}}$ matrix collecting the values of the $\tilde{d}$ external features over $\tilde{p}$ time steps
- $\hat{X}_{t+1:t+H} \in \mathbb{R}^{H \times d}$ matrix of all d time series forecasts for the periods $t + 1, t + 2, \dots, t + H$

■ Be aware of terminological pitfalls and differences across fields

Generalized Time Series Forecasting Models

A single architecture supports all global model types shown above

- **Need input and output layers of suitable sizes**
- **Need to arrange the data suitably**
 - Note the inaccuracy in the visualization
 - X is a matrix. Thus, we need multiple input nodes to feed the network all its data
 - We can stack the values of features over past time steps on top of each other ($\rightarrow$ flatten the matrix)
 - Same for the output layer
- **Need to ensure correct loss calculation**
 - Align $\hat{X}_{t+1:t+H}$ and $X_{t+1:t+H}$
 - Could also weight losses
 - Higher weight to more recent forecasts
 - Higher weight to more important time series



Take a Breath

■ Time series forecasting is a broad field requiring a flexible modeling framework

- Model input
 - One/multiple series of past realizations of the target/s
 - Potentially further explanatory/external time series
- Model output
 - Forecasts of the/all time series
 - Potentially over multiple future periods

■ Feedforward nets support time series forecasting

- Straightforward to adjust the network architecture
- Loss calculations may also need adjustment
- Data must be arranged properly into vectors/matrices

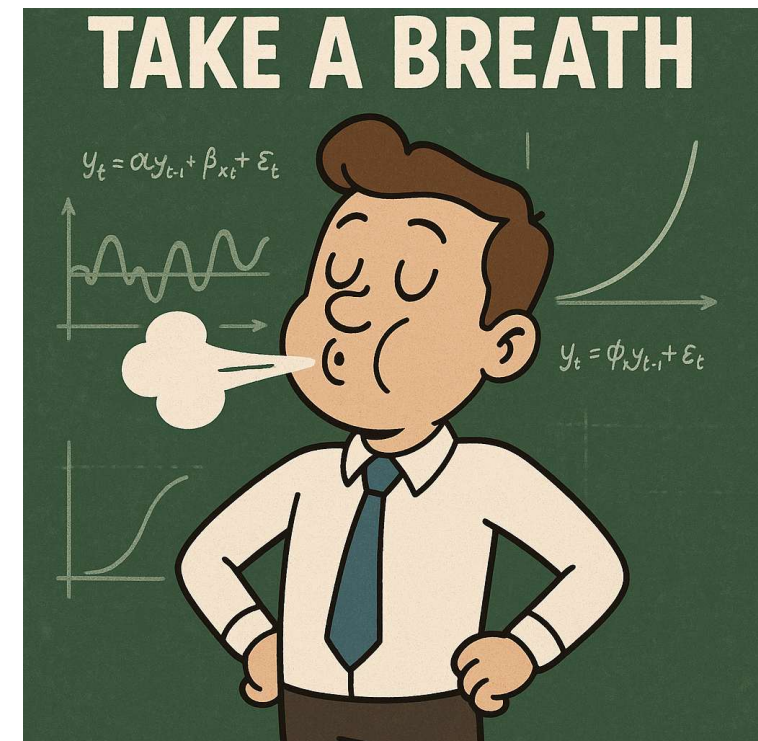
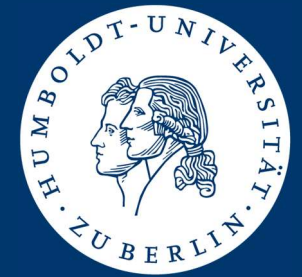


Image by ChatGPT4o

Prompt: „Generate a funny picture for a presentation slide with the title "Take a breath".
Context: the presentation walked students through different types of time series forecasting models and their mathematical underpinnings. Hence, time to take a breath.

Commonalities Across Discussed Model Types and Remaining Gaps

- All shown approaches represent **autoregressive (AR) models**. Their core idea is to predict based on a fixed-sized window of past realizations of the time series
- All shown approaches provide **point forecasts**. We can interpret a forecast $\hat{x}_t$ as the conditional expectation of the value at time point t given the past values
- **Relevant extensions for NLP**
 - Probabilistic forecasting: instead of predicting a point estimate, $\hat{x}_{t+h}$, predict the distribution over possible values, $\hat{x}_{t+h} \sim p(x_{t+h} | x_{t-p+1:t})$
 - Address limitations of the AR modeling approach
 - Fixed size input window
 - Cannot adapt to variable lag structures
 - Capturing long-term dependencies is a problem (e.g., very long input sequences, high dimensionality, etc)
 - No internal memory
 - Past values are treated as independent features
 - Difficult to capture temporal dynamics (e.g., trend, seasonality, regime changes, etc.)
 - Poor extrapolation



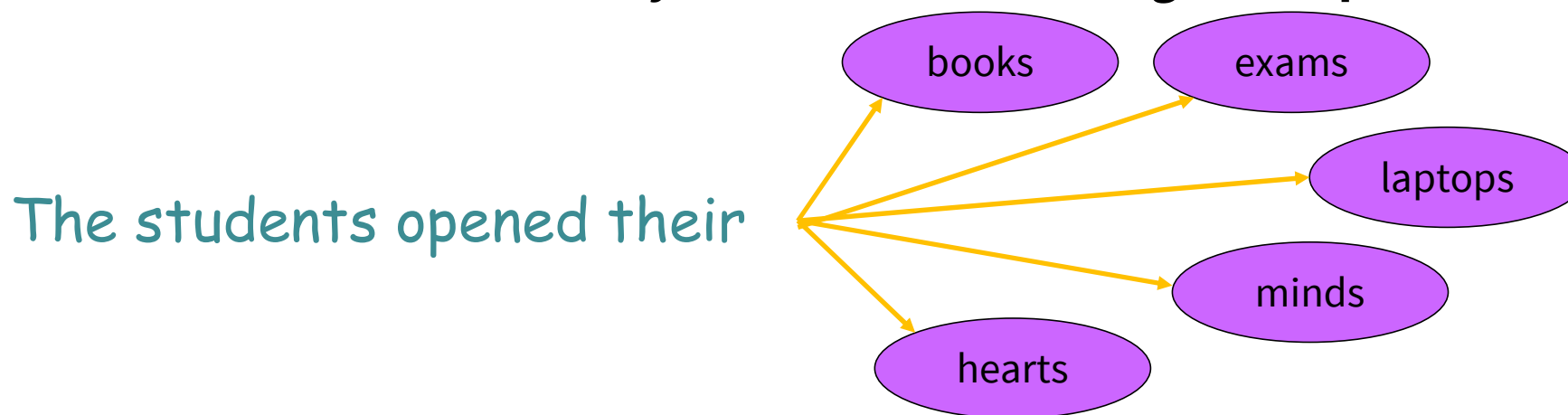
Language Models

Definition & purpose, probabilistic forecasting, requirements

Language Models (LM)

A LM predict how an input sequence of tokens (e.g., words) continues

- Given a sequence of tokens $x_1, x_2, \dots, x_t$
- Where x_t can be any token out of a fixed vocabulary $V = \{v_1, v_2, \dots, v_{|V|}\}$
- Compute a **probability distribution** over tokens
- To determine the most likely next token continuing the sequence



$$\hat{x}_{t+1} \sim p(x_{t+1} | x_t, x_{t-1}, \dots, x_1)$$

Why Talking About Language Models?

Language modeling is a pivotal NLP task

■ LMs have many direct applications

- Auto-complete function
- Messengers, search engines, text processors, etc.

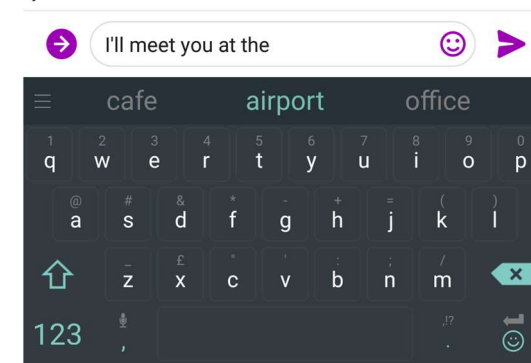
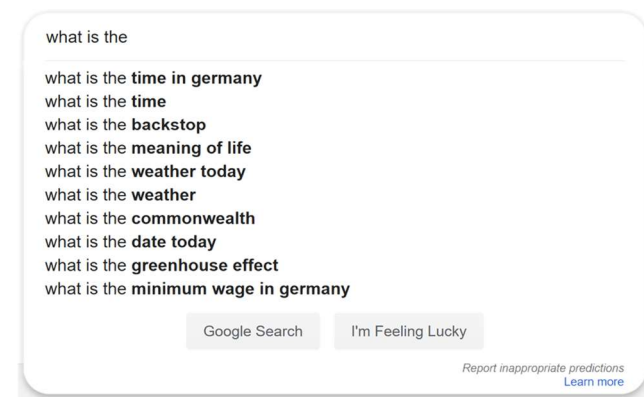
■ Standardized benchmarks to measure progress in NLP

■ Language modeling turned out to be a crucial **upstream** NLP task

- Many downstream apps make use of language models (machine translation, text summarization)
- Pre-training and transfer learning rely on LMs (see later)

■ State-of-the-art AIs are based on large LMs

- Yann LeCun expressed concern that auto-regressive nature of today's LLMs is a bottleneck hindering further AI progress
- [\(6\) Post | LinkedIn](#)
- [Meta AI chief says large language models will not reach human intelligence](#)



Language Modeling involves Probabilistic Forecasting

A system that assigns probability to a piece of text

■ Goal of probabilistic forecasting

- Instead of predicting a **point estimate**, predict a **probability distribution** over possible values
- $\hat{x}_{t+h} = f_w(x_{t-p+1:t})$ versus $\hat{x}_{t+h} \sim p(x_{t+h} | x_{t-p+1:t})$
- We can setup all forecasting model types previously discussed as probabilistic models
 - Let network outputs represent the parameters of a continuous distribution
 - Example Gaussian distribution: the network models $\mu(x)$ and $\sigma(x)$ using two output nodes
 - We can then sample $\hat{x}_{t+1} \sim N(\mu(x_{t-p:t}), \sigma(x_{t-p:t})^2)$

■ (Text-only) LLMs are univariate, auto-regressive, probabilistic single output models

■ Formalization of the modeling task

- Probability of observing a sequence of tokens $x_1, x_2, \dots, x_T$
- $p(x_1, x_2, \dots, x_T) = p(x_1) \times p(x_2 | x_1) \cdot \dots \cdot p(x_T | x_{T-1}, \dots, x_1) = \prod_{t=1}^T p(x_t | x_{t-1}, \dots, x_1)$

A Neural Language Model

Using a neural network with fixed window to build a language model

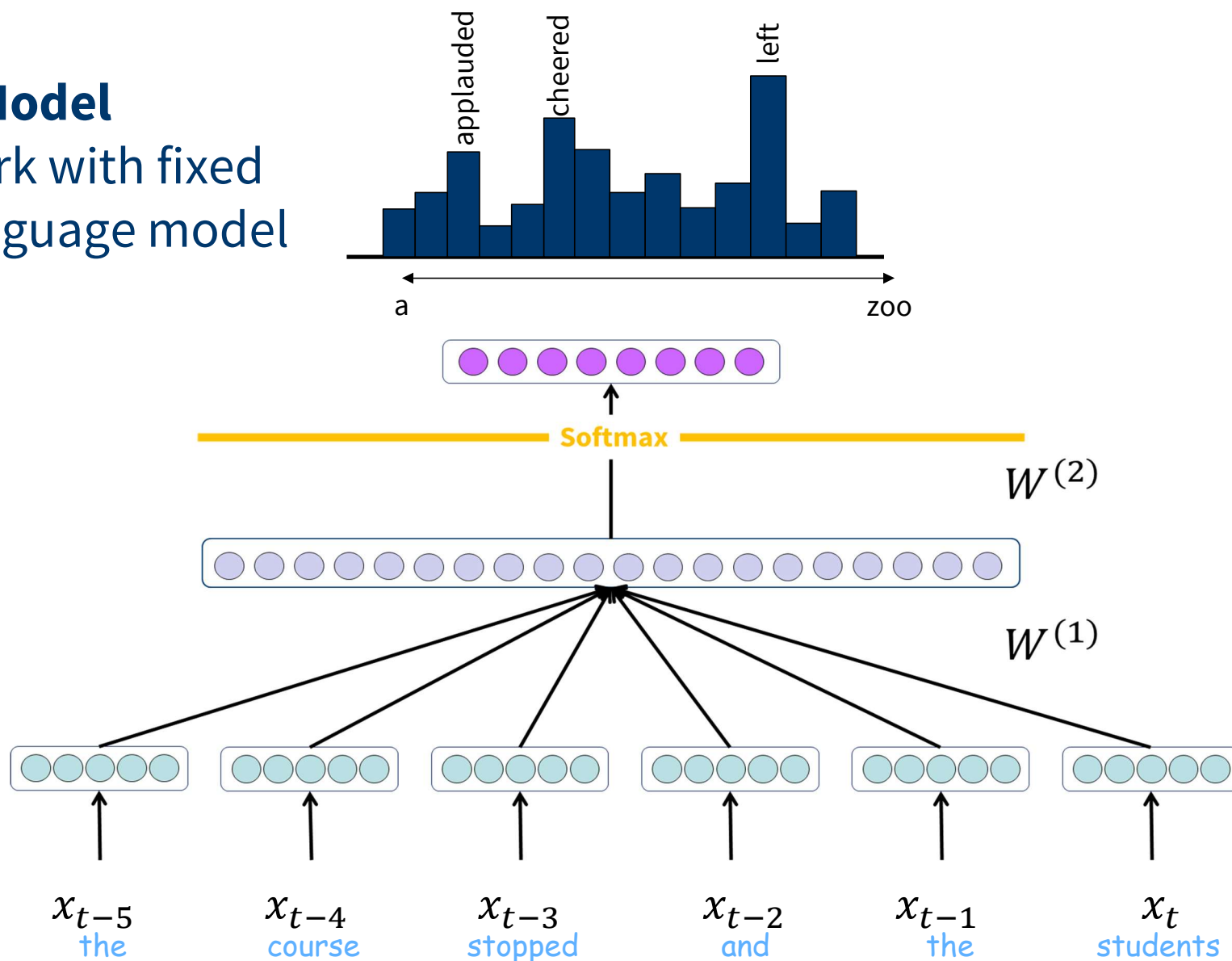
Output: probability distribution of the next word in the sequence

$$p(x_{t+1}|x_t, \dots, x_1)$$

Hidden layer

Word Embeddings
(e.g., W2V)

Input: sequence of tokens
 $x_1, x_2, \dots, x_t$



Neural Language Model Problems

Problems with long term dependencies and lack of parameter sharing

■ Window size typically short

- We might need information from the distant past to understand text
- Larger windows amplify no. of parameters and could lead to many irrelevant inputs

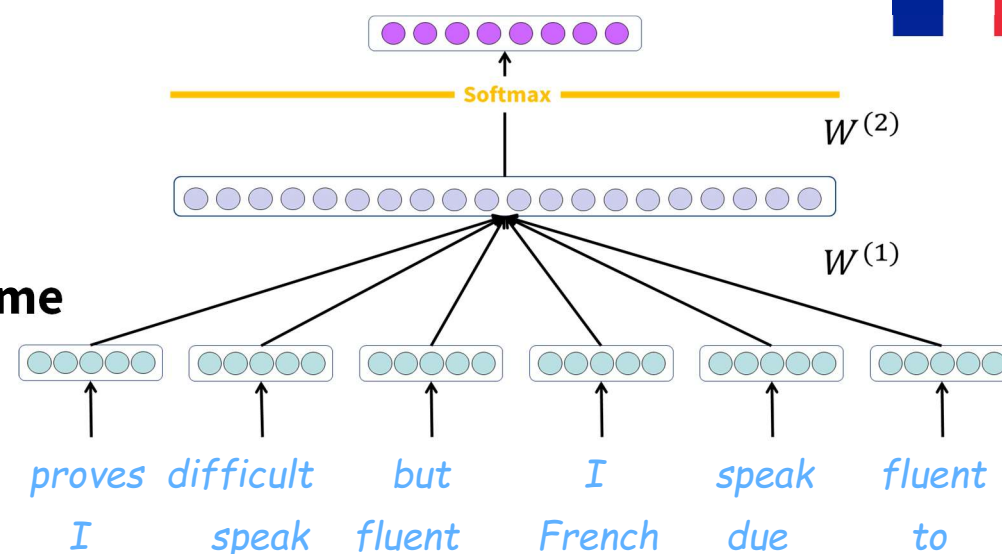
■ Each input (word) has separate parameters

- The same sequence might appear at different positions in a text
- Information once learnt about a sequence will not transfer
- No proper notion of word order

■ The dimensionality of the input grows over time

- Two previous tokens to predict the third token,
- Three prev. tokens to predict token fourth, etc.
- $p(x_1, x_2, \dots, x_T) = \prod_{t=1}^T p(x_t | x_{t-1}, \dots, x_1)$

I grew up in France where my father works as a teacher in primary school. My mom is a criminal lawyer. Just recently, I moved to Berlin for my undergraduate studies. German proves difficult but I speak fluent ???.



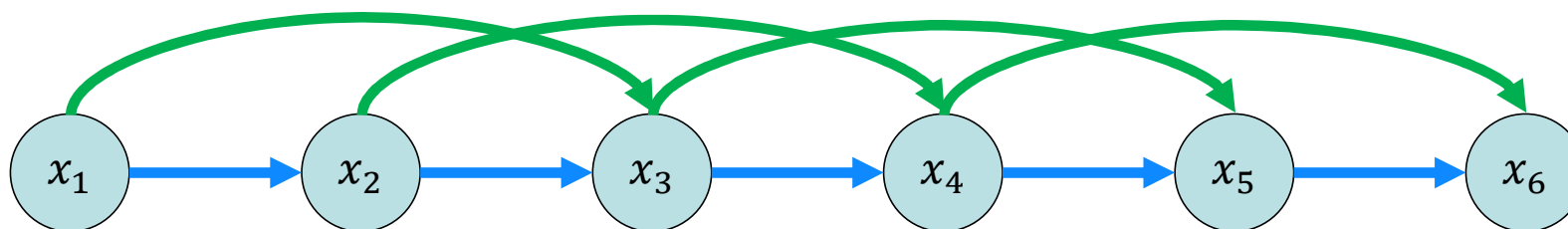
Statistical Tools for Sequential Data

Plan A: Markov Assumption

■ Maybe we do not need all past observations $x_{t-1}, x_{t-2}, \dots, x_1$

- Using only the last τ observations ensures fixed input length for all $t > \tau$
- Parameter τ represents a believe of the dependence structure

$$p(x) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1, x_2) \cdot \dots \cdot p(x_t|x_{t-\tau}, \dots, x_{t-1})$$



First order Markov model

Second order Markov model

Statistical Tools for Sequential Data

Plan B: Latent variable or state space model

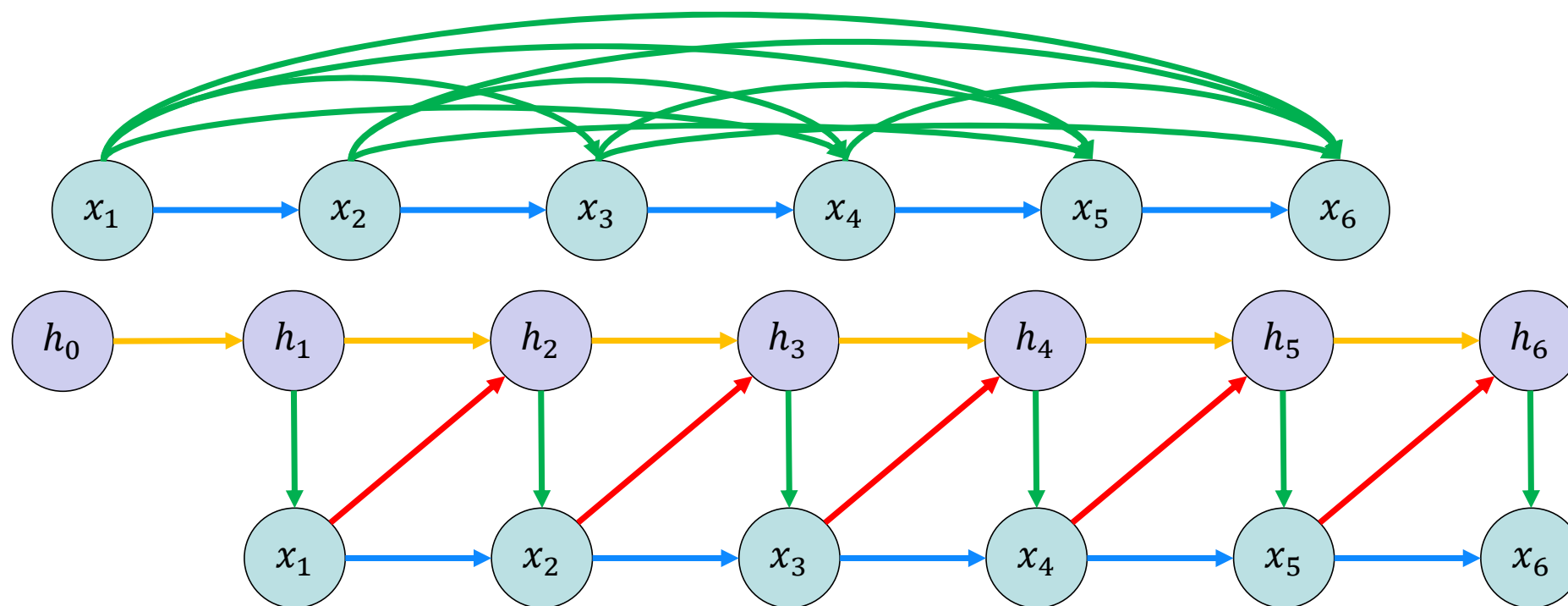
- Keep some memory h_t of past observations

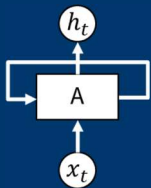
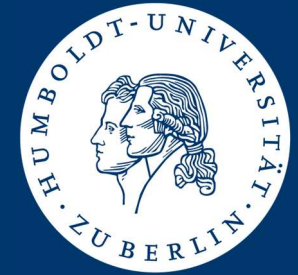
- Update the memory when new information arrives

- Note that state-space models encompass AR, MA, ARMA, and ARIMA models as special cases

$$p(x_t | \underline{h_t}, \underline{x_{t-1}})$$

$$p(h_t | \underline{h_{t-1}}, \underline{x_{t-1}})$$





Recurrent Neural Networks (RNN)

Recurrent cells, computational graph, and language modeling examples

Recurrent Versus Feedforward Neural Networks

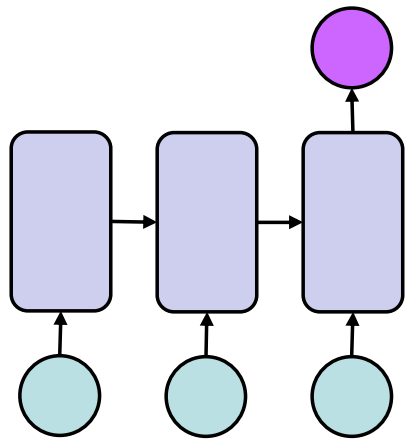
■ Feedforward networks

- Input to output mapping using vectors of fixed size
- Fixed number of computational steps (e.g., hidden layer(s))

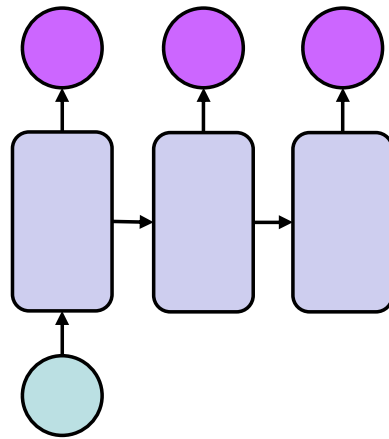


■ Recurrent networks provide a flexible sequence modeling framework

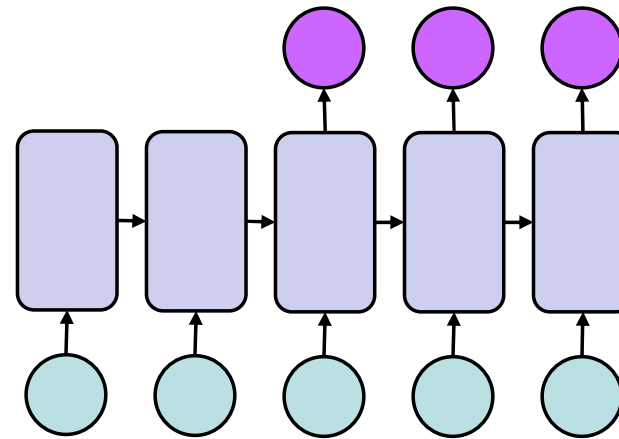
- Operate over sequences of vectors
- Sequences in the input layer, in the output layer, or in both layers



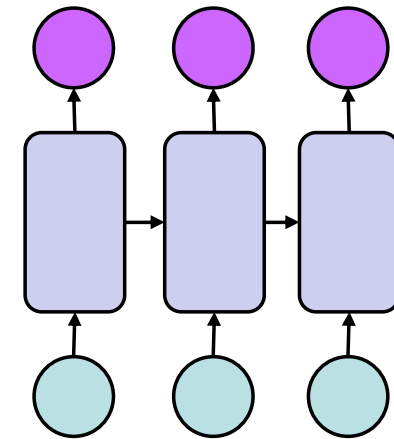
Many to one



One to many



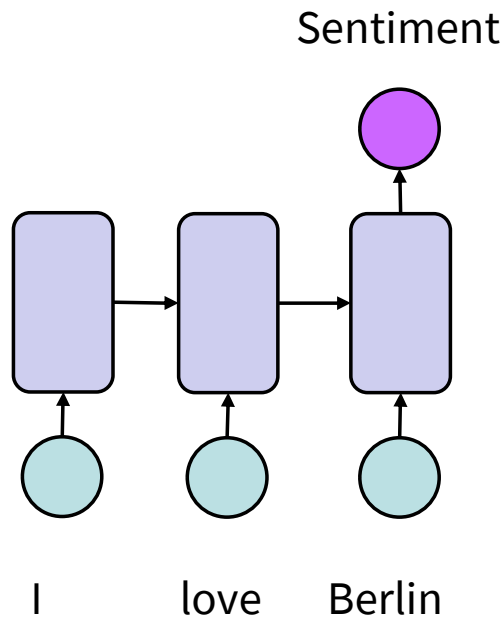
Many to many without sync.



Synced many to many

Recurrent Versus Feedforward Neural Networks

Exemplary application: sentiment analysis



Susan Athey @Susan_Athey · Apr 20

Pathbreaking work that also demonstrates the power of using administrative data for actionable research. [@JHastings_Econ](#)

Life Interventions for Very Low Birth Weight Newborns
Chyn, Samantha Gold, Justine S. Han
NBER Working Paper No. 25753
Issued in April 2019
Journal of Human Capital, Health Care, Health Economics
Administrative data from Rhode Island to measure the impact of early-life interventions for low birth weight newborns on later-life outcomes. We use a 3-year threshold for Very Low Birth Weight (VLBW) more intense in-hospital care, in line with prior studies, to increase in test scores in elementary school, the probability of college enrollment, and \$66,997 explore potential mechanisms driving these impacts

NBER @nberpubs

Using administrative data from Rhode Island to measure the impact of early-life interventions for low birth weight newborns on later-life outcomes, from Eric Chyn, Samantha Gold, and [@JHastings_Econ](#) [nber.org/papers/w25753](#)



3



33



146



Kendra Stenberg @StenbergKendra · Apr 24

Ughh, train is delayed 🤬



Gloria Yoyis Sotelo @gloyoyissotes · Apr 24

What usually takes a few minutes to find parking today took me 20 minutes. Coming to the last week of school isn't going to save your semester

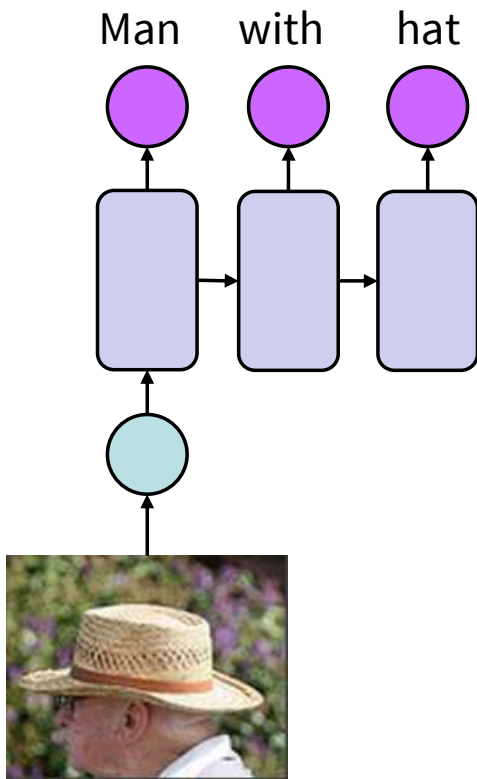


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Recurrent Versus Feedforward Neural Networks

Exemplary application: image captioning



"man in black shirt is playing guitar."



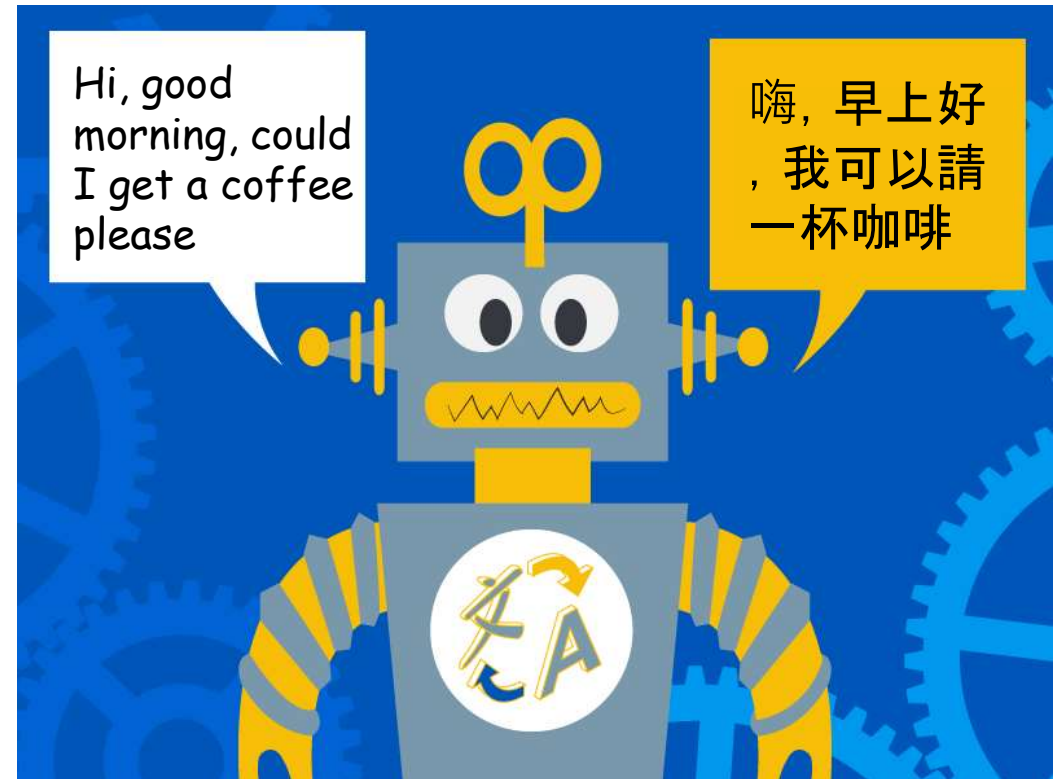
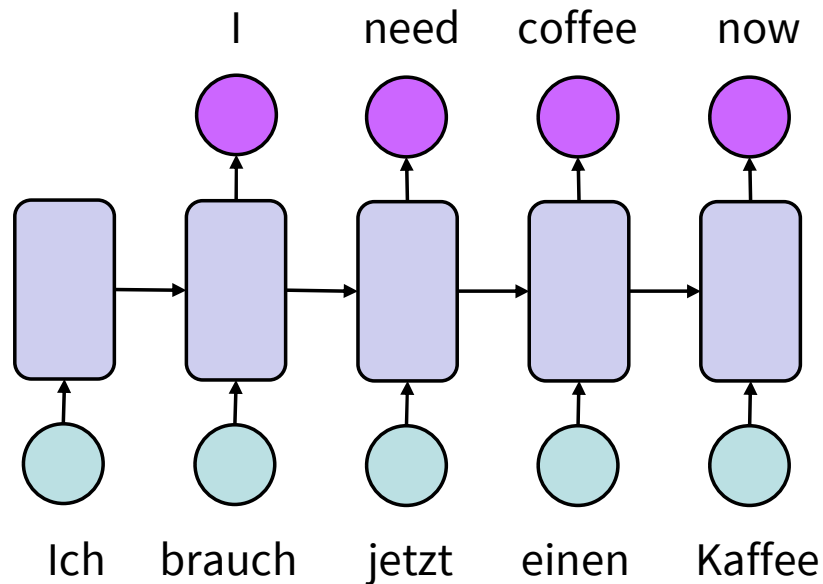
"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."

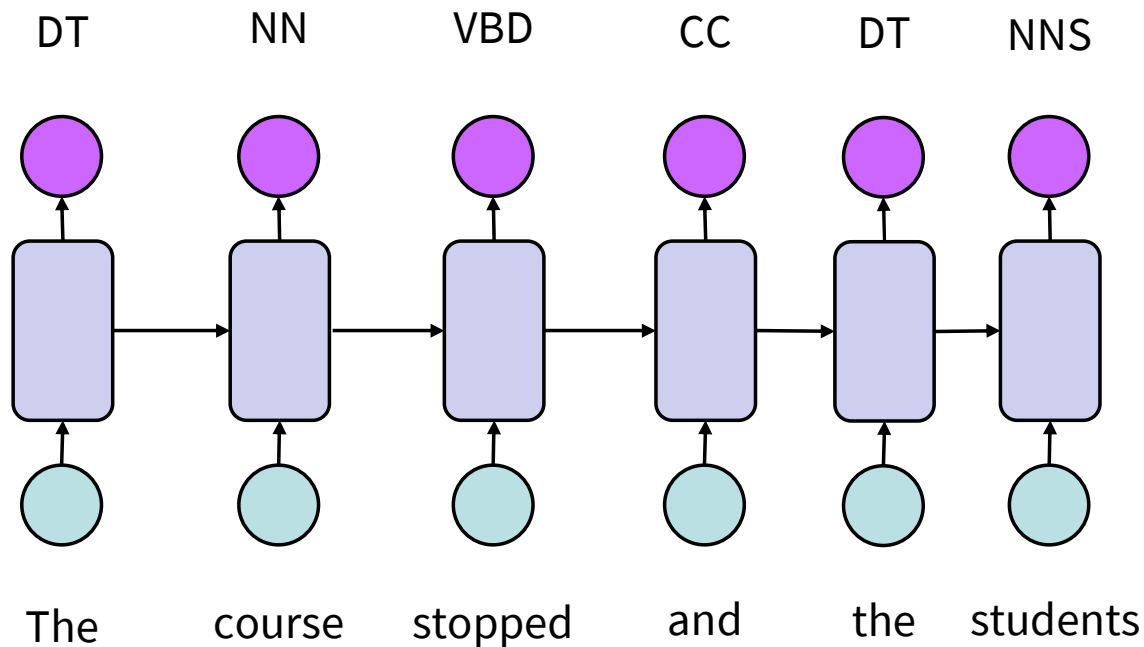
Recurrent Versus Feedforward Neural Networks

Exemplary application: machine translation



Recurrent Versus Feedforward Neural Networks

Exemplary application: part of speech tagging

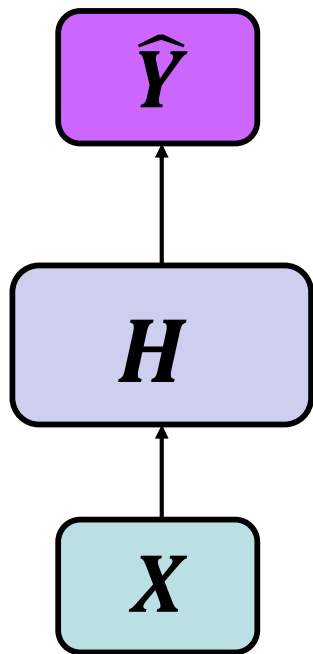


Brown/Penn Treebank tags

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	<i>and, but, or</i>	SYM	Symbol	<i>+, %, &</i>
CD	Cardinal number	<i>one, two, three</i>	TO	"to"	<i>to</i>
DT	Determiner	<i>a, the</i>	UH	Interjection	<i>ah, oops</i>
EX	Existential 'there'	<i>there</i>	VB	Verb, base form	<i>eat</i>
FW	Foreign word	<i>mea culpa</i>	VBD	Verb, past tense	<i>ate</i>
IN	Preposition/sub-conj	<i>of, in, by</i>	VBG	Verb, gerund	<i>eating</i>
JJ	Adjective	<i>yellow</i>	VBN	Verb, past participle	<i>eaten</i>
JJR	Adj., comparative	<i>bigger</i>	VBP	Verb, non-3sg pres	<i>eat</i>
JJS	Adj., superlative	<i>wildest</i>	VBZ	Verb, 3sg pres	<i>eats</i>
LS	List item marker	<i>1, 2, One</i>	WDT	Wh-determiner	<i>which, that</i>
MD	Modal	<i>can, should</i>	WP	Wh-pronoun	<i>what, who</i>
NN	Noun, sing. or mass	<i>llama</i>	WP\$	Possessive wh-	<i>whose</i>
NNS	Noun, plural	<i>llamas</i>	WRB	Wh-adverb	<i>how, where</i>
NNP	Proper noun, singular	<i>IBM</i>	\$	Dollar sign	<i>\$</i>
NNPS	Proper noun, plural	<i>Carolinas</i>	#	Pound sign	<i>#</i>
PDT	Predeterminer	<i>all, both</i>	"	Left quote	<i>(' or ")</i>
POS	Possessive ending	<i>'s</i>	"	Right quote	<i>(' or ")</i>
PP	Personal pronoun	<i>I, you, he</i>	(	Left parenthesis	<i>(, (, {, <)</i>
PP\$	Possessive pronoun	<i>your, one's</i>	)	Right parenthesis	<i>(,), }, >)</i>
RB	Adverb	<i>quickly, never</i>	,	Comma	<i>,</i>
RBR	Adverb, comparative	<i>faster</i>	.	Sentence-final punc	<i>(. ! ?)</i>
RBS	Adverb, superlative	<i>fastest</i>	:	Mid-sentence punc	<i>(: ; ... - -)</i>
RP	Particle	<i>up, off</i>			

Feedforward Neural Network Revisited

Hidden layer has no memory and considers each case individually



$$\hat{Y} = HW_{hy} + b_y$$

$$H = g(XW_{xh} + b_h)$$

Notation

$$X \in \mathbb{R}^{n \times m}$$

$$W_{xh} \in \mathbb{R}^{m \times h}$$

$$b_h \in \mathbb{R}^{1 \times h}$$

$$H \in \mathbb{R}^{n \times h}$$

$$W_{hy} \in \mathbb{R}^{h \times c}$$

$$b_y \in \mathbb{R}^{1 \times c}$$

$g(\cdot)$ activation fun.

n No. of cases; e.g., minibatch size

m No. of input variables

c No. of classes / neurons in output layer

h No. of hidden units

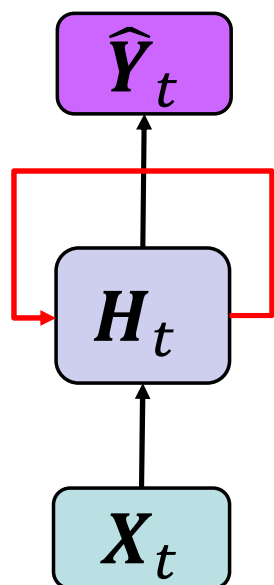
A Recurrent Neural Network

The recurrent cell accumulates information over multiple time steps

■ Recurrent cell accumulates information over multiple time steps

■ Latent variable or state-space model

- Nonlinear, learnable state-space model
- State update and output function are (in a sense) neural networks



$$\hat{\mathbf{Y}}_t = \mathbf{H}_t \mathbf{W}_{hy} + \mathbf{b}_y$$

$$\mathbf{H}_t = g(\mathbf{X}_t \mathbf{W}_{xh} + \mathbf{H}_{t-1} \mathbf{W}_{hh} + \mathbf{b}_h)$$

Notation:

$$\mathbf{X}_t \in \mathbb{R}^{n \times m}$$

$$\mathbf{W}_{xh} \in \mathbb{R}^{m \times h}$$

$$\mathbf{b}_h \in \mathbb{R}^{1 \times h}$$

$$\mathbf{H}_t \in \mathbb{R}^{n \times h}$$

$$\mathbf{W}_{hh} \in \mathbb{R}^{h \times h}$$

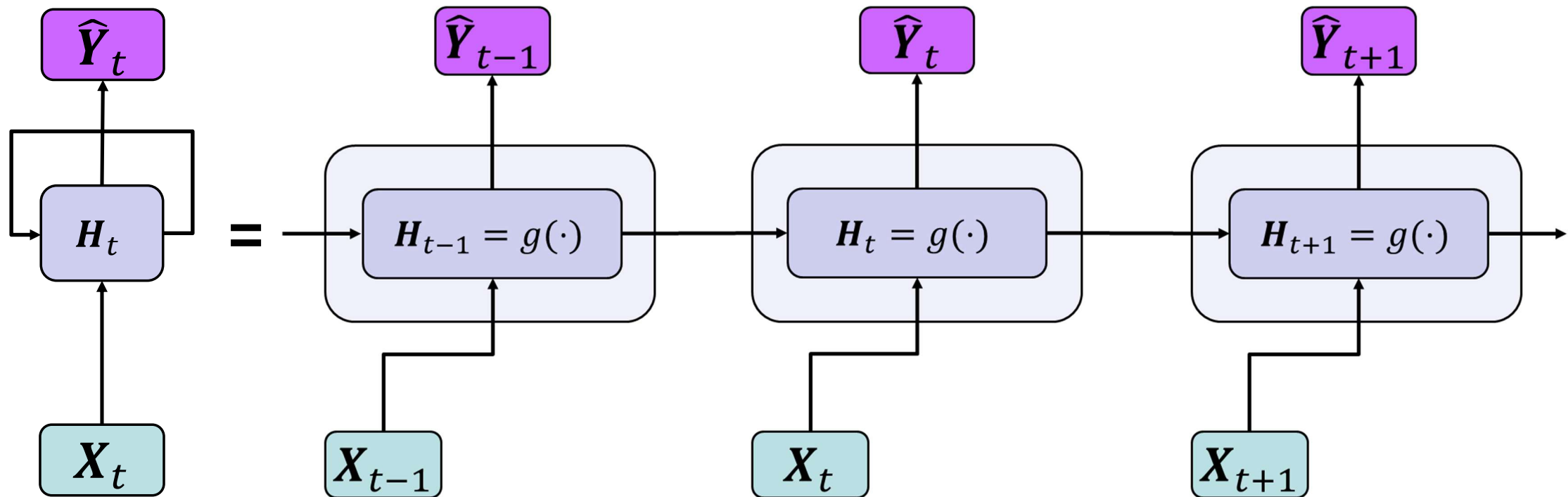
$$\mathbf{W}_{hy} \in \mathbb{R}^{h \times c}$$

$$\mathbf{b}_y \in \mathbb{R}^{1 \times c}$$

$g(\cdot)$ activation fun.

A Recurrent Neural Network Unfolded in Time

Illustration of forward calculations across three time steps



An RNN Language Model

Output distribution

$$\hat{x}_{t+1} \in (0,1)^{|V|} = \text{softmax}(W_{hy}H_t + b_y)$$

hidden state

$$H_t = g(e_1W_{eh} + H_{t-1}W_{hh} + b_h)$$

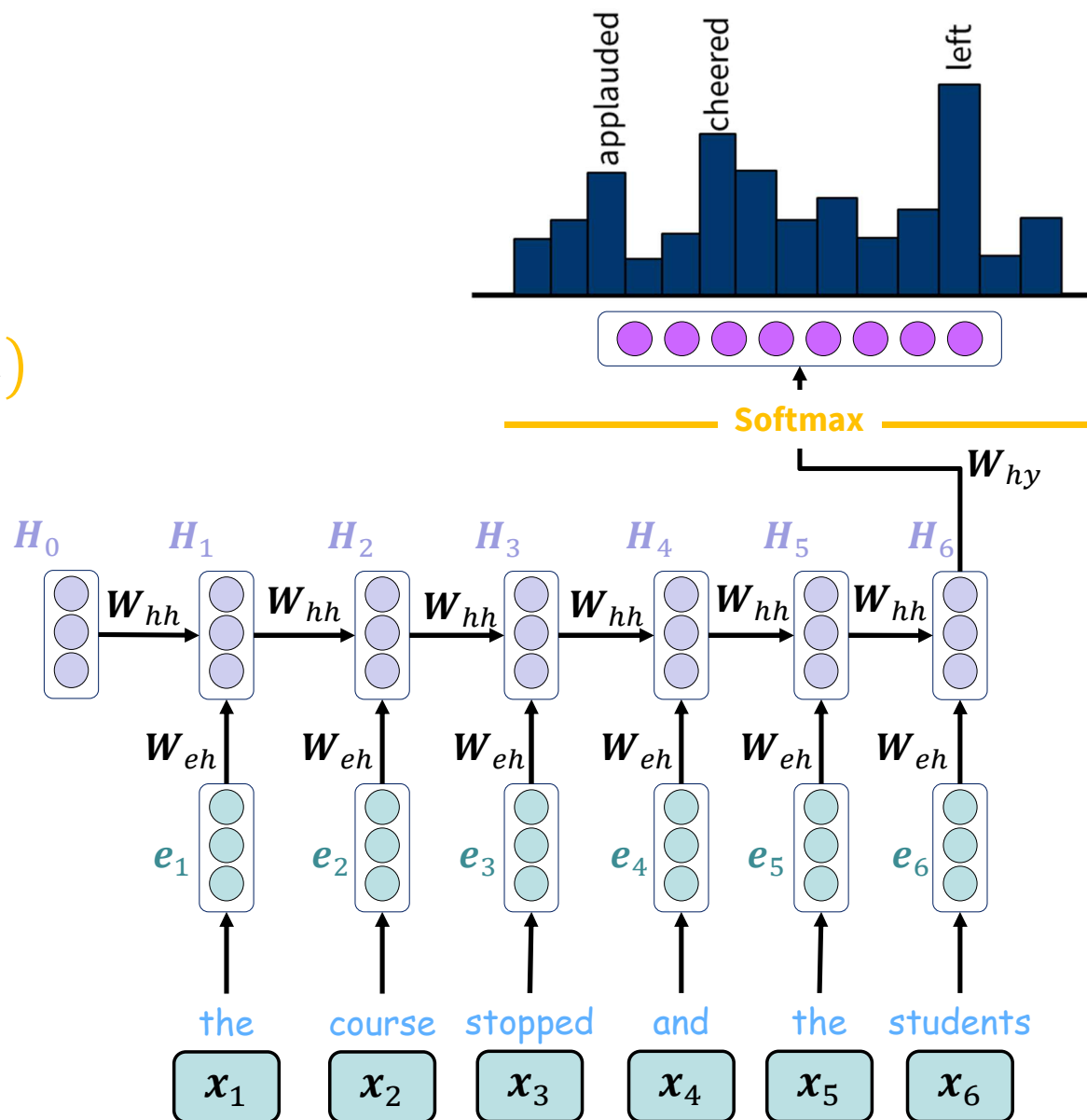
H_0 = is the initial hidden state

word embedding

$e_t \in \mathbb{R}^d = Ex_t$, with $E \in \mathbb{R}^{d \times |V|}$
with embedding dimensions d

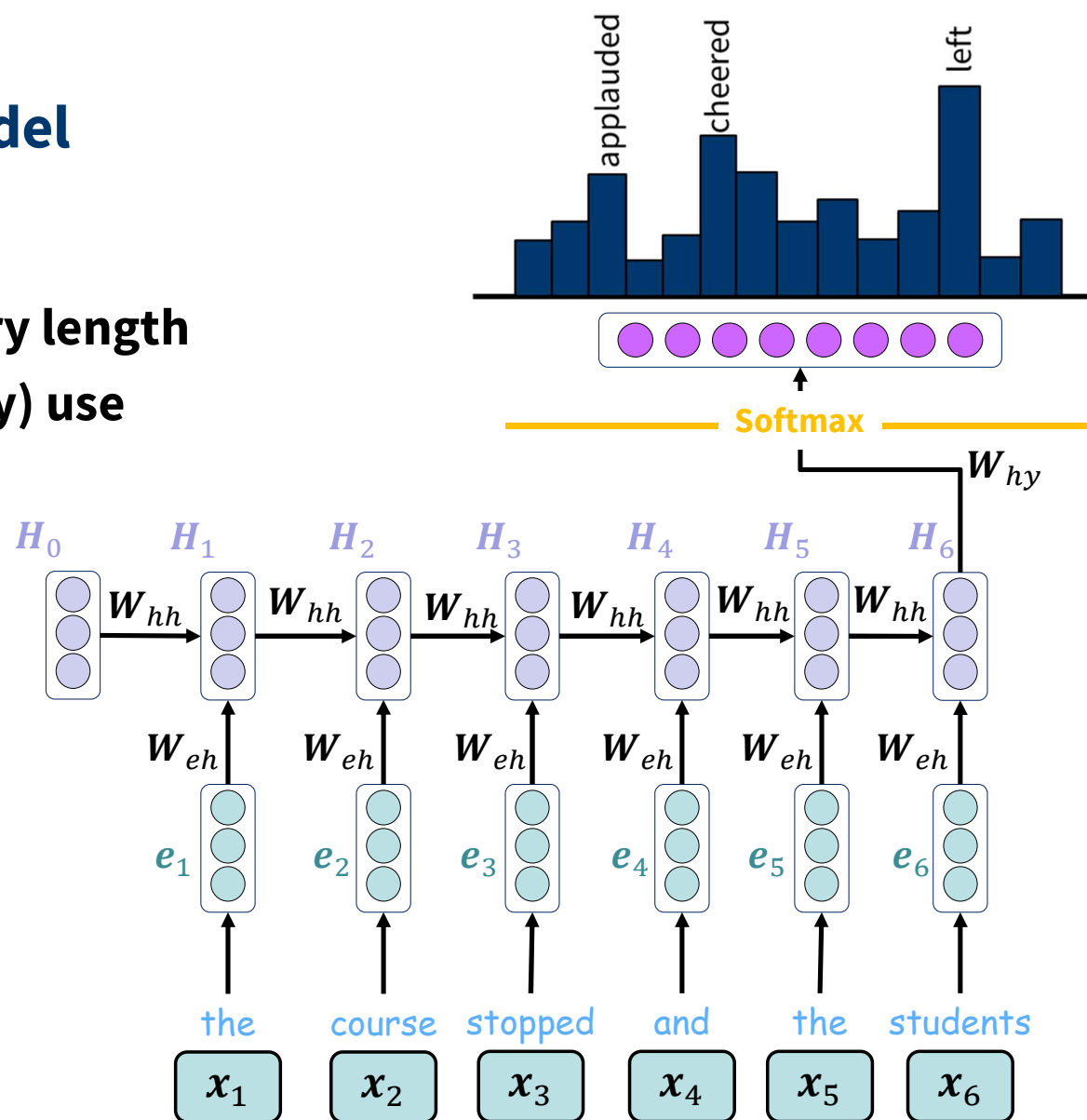
word as one-hot vector

$$x_t \in \{0,1\}^{|V|}$$



Advantages of RNN Language Model

- Supports input sequences of arbitrary length
- Computation for a step can (in theory) use information from many steps back
- Model size does not increase for longer input
- Same weight applied on every time step, so there is symmetry in how inputs are processed
- Caveats
 - Recurrent computation is slow
 - In practice, difficult to access information from many steps back



Generating Text with an RNN-LM

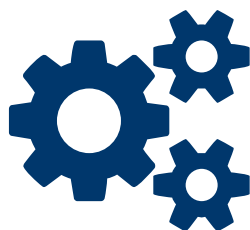
Use 1-step forecast recursively

- Input sequence with predefined length to accumulate hidden state
- Apply output-weight matrix, to obtain raw output scores
- Softmax raw scores to obtain probability distribution
- Select most probable next token as prediction
- Shift input window and repeat

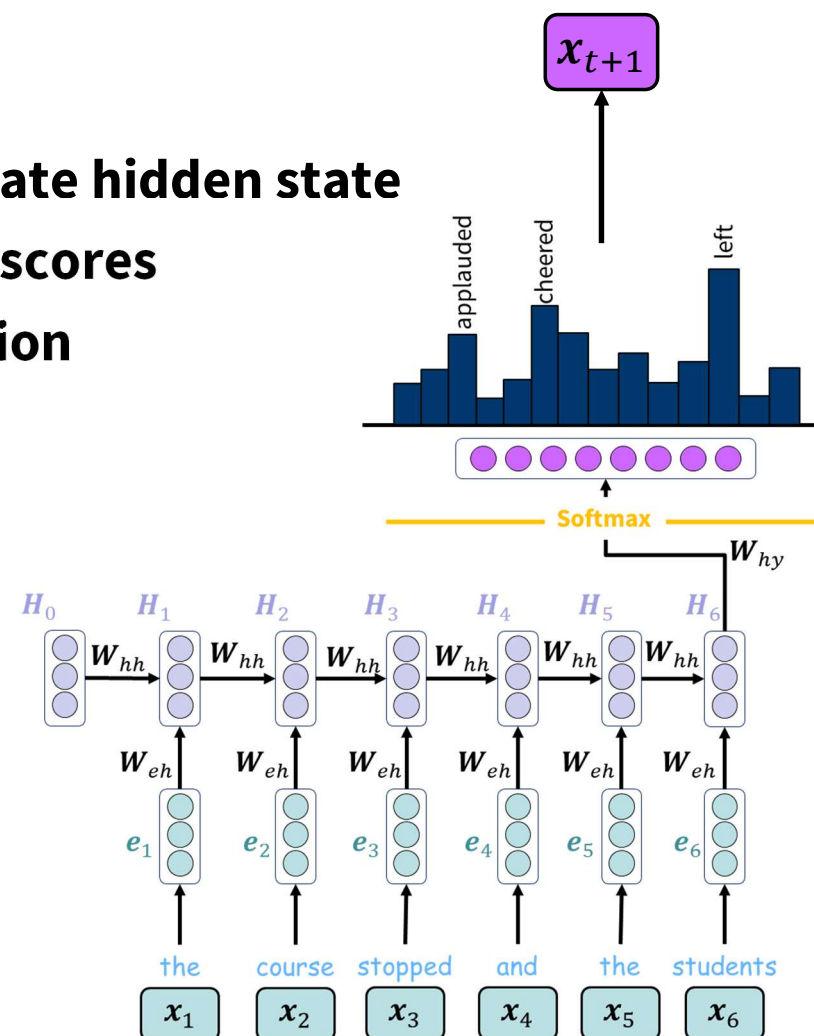
Corpus



Sampling



started as early as eight in
in the morning <comma> talking about
about RNNs <dot> After a daunting
daunting two hours it eventually turned
eventually turned ten and the course
the course stopped and the students
students closed their books and left



Generating Text with RNN Language Model

RNN-LM trained on Obama speeches



The United States will step up to the cost of a new challenges of the American people that will share the fact that we created the problem. They were attacked and so that they have to say that all the task of the final days of war that I will not be able to get this done. The promise of the men and women who were still going to take out the fact that the American people have fought to make sure that they have to be able to protect our part. It was a chance to stand together to completely look for the commitment to borrow from the American people. And the fact is the men and women in uniform and the millions of our country with the law system that we should be a strong stretchs of the forces that we can afford to increase our spirit of the American people and the leadership of our country who are on the Internet of American lives.

Thank you very much. God bless you, and God bless the United States of America.

Generating Text with RNN Language Model

RNN-LM trained on Harry Potter I - IV



"The Malfoys!" said Hermione.

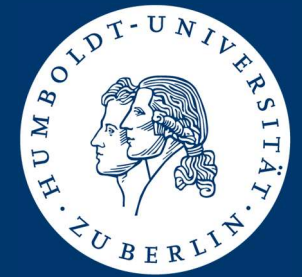
Harry was watching him. He looked like Madame Maxime. When she strode up the wrong staircase to visit himself.

"I'm afraid I've definitely been suspended from power, no chance—indeed?" said Snape. He put his head back behind them and read groups as they crossed a corner and fluttered down onto their ink lamp, and picked up his spoon. The doorbell rang. It was a lot cleaner down in London.

Hermione yelled. The party must be thrown by Krum, of course.

Harry collected fingers once more, with Malfoy. "Why, didn't she never tell me. ..." She vanished. And then, Ron, Harry noticed, was nearly right.

"Now, be off," said Sirius, "I can't trace a new voice."



Summary

Summary



Learning goals

- Forms of time series forecasting
- LM applications and formalization
- RNN neural networks



Findings

- NNs are AR models and facilitate forecasting
- LMs are univariate, probabilistic forecasting models
- RNNs maintain an internal memory (state)
- Incremental updates to that memory (hidden state) accumulate information over time
- RNNs provide a better framework for LM than feedforward networks



What next

- RNN model training
- Problems with vanilla RNNs (vanishing gradient)

Credits & Disclaimer

- **Course materials have been inspired and draw upon numerous great resources available online**
- **The authors acknowledge the use of and wish to express their gratitude to the following sources**
 - Colah's blog, Understanding LSTM networks (<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>)
 - Zhang et al.: Dive into Deep Learning (<http://d2l.ai/index.html>)
 - MIT 6.S191: Deep Sequence Modeling (<http://introtodeeplearning.com/>)
 - Stanford University: Natural Language Processing with Deep Learning (<https://web.stanford.edu/class/cs224n/>)
 - Oxford University: Deep Natural Language Processing course (<https://github.com/oxford-cs-deepnlp-2017>)

Thank you for your attention!

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